Renaldo Miya

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EDUCATION

Bachelor of Mechanical Engineering **Expected Graduation: April 2029**
University of Waterloo Waterloo, ON

- UW Formula Electric — Mechanical Design & Composites Member, 2024

WORK EXPERIENCE

Engineering Design Student **September 2026 – December 2026**
Amphenol Markham, ON

- Model and draft connector and motor jack components for aerospace and commercial systems in SolidWorks, producing detail and assembly drawings for manufacturing
- Support new product design from concept CAD through design review, revising geometry against manufacturability and assembly feedback
- Update models, drawings, and documentation through engineering change notices on released products, maintaining configuration control across revisions
- Support design testing and qualification of connector assemblies, investigating optimization and safety issues and feeding findings back into design revisions

ARIEL Engineering Assistant **January 2026 – April 2026**
TRIUMF | Canada's national particle accelerator laboratory Vancouver, BC

- Designed a compact stainless-steel support frame in SolidWorks for a radioactive gas pumping manifold, forming a rigid, wall-mounted enclosure with minimal floor footprint
- Sized members and clamp layout through iterative ANSYS FEA, holding deflection under 2 mm and stress under 105 MPa while minimizing part count in a radiation-exposed assembly
- Produced ASME Y14.5 GD&T manufacturing drawings for fabrication and inspection, reducing fastener count and streamlining assembly in the ARIEL Target Hall
- Supported fabrication on manual machining tools, validating tolerances and resolving fit-up issues to translate CAD design intent into the installed structure

Mechanical Engineering Co-op **May 2024 – August 2024**
BC Hydro | Site C Clean Energy Project Fort St John, BC

- Reviewed and redlined P&IDs and mechanical layouts against applicable codes and standards, flagging constructability issues and issuing modifications adopted into the build
- Conducted on-site field inspections to verify installation accuracy and code compliance, documenting discrepancies and driving corrective design changes that reduced rework

PROJECTS

Rack-and-Pinion Gripper | UW Robotics Design Team **Fall 2025**

- Designed and built a motor-driven gripper with a 5 kg lift capacity, owning it from requirements through functional prototype
- Sized the rack-and-pinion drivetrain converting motor rotation to linear jaw travel, selecting gear ratio and jaw geometry to meet grip force and stroke targets
- Combined 3D-printed structure with machined aluminium load-path parts, trading mass and manufacturability against stiffness under peak grip load
- Now developing a v2 with closed-loop grip force control and a characterization plan for force accuracy, slip resistance, and durability cycling

TECHNICAL SKILLS

CAD & Modelling | SolidWorks, AutoCAD, DraftSight (2D)
Analysis & Simulation | ANSYS, SolidWorks Simulations, MATLAB
Manufacturing| Mill, Lathe, Drill Press, 3D Printing, Composite Layup
Design & Documentation | GD&T (ASME Y14.5), DFM/DFA, BOM Management, SolidWorks PDM, Windchill, P&ID
Programming | C++, C, Python, HTML/CSS